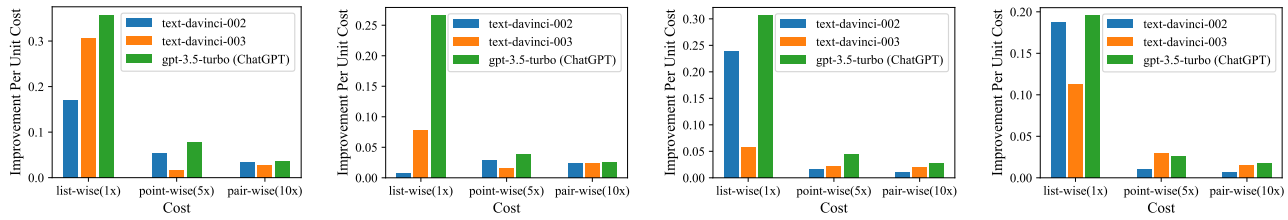
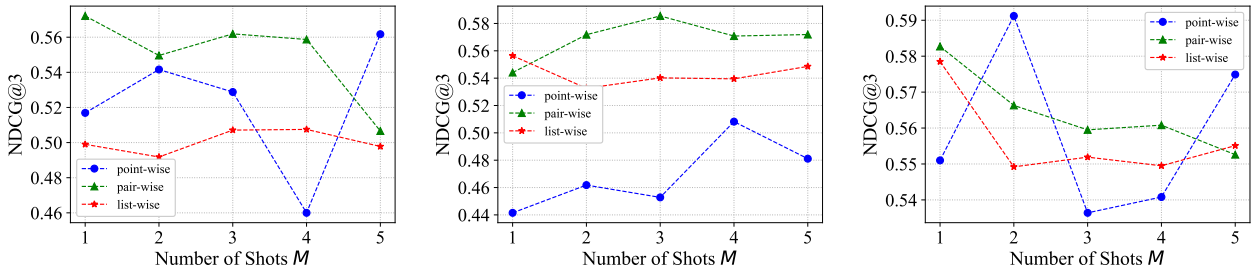


(a) Movie (b) Book (c) Music (d) News

Figure 3: Improvement of $NDCG@3$ per unit cost and five shots examples on four datasets. '1x 5x 10x' denote the cost of list-wise, point-wise, and pair-wise, respectively.



(a) text-davinci-002 (b) text-davinci-003 (c) gpt-3.5-turbo (ChatGPT)

Figure 4: Impact of the number of shots prompts in LLMs on Movie dataset.

models do not require training data but rather a few samples in the prompt to help understand the recommendation task. Therefore, we conclude that LLM-based recommendation models can be applied in practice to mitigate the cold start problem.

4.4 RQ3: Performance Scaling by Cost

Although the LLMs have better performance on pair-wise or list-wise ranking as presented in Table 1, we need to consider the costs associated with these performance improvements. Specifically, we calculate the improvement per unit cost for each LLM: $\frac{\frac{V_{LLM}-V_{random}}{V_{random}}}{cost_{LLM}}$, where V_{LLM} denotes the metric value of the LLM, V_{random} denotes the metric value of random recommendation, $cost_{LLM}$ denotes the cost of ranking one user's candidate item list. Referring to Figure 1 (left), let us define the $cost_{LLM}$. For list-wise ranking, only one prompt input is needed to obtain LLM's ranking for all candidate items. For point-wise ranking, N prompt inputs are required to obtain LLM's ranking for all candidate items (where N is the number of candidate items). For pair-wise ranking, $\frac{N(N-1)}{2}$ prompt inputs are required to obtain all ranking results. In our experimental settings, N is set to 5. The costs of point-wise, pair-wise, and list-wise ranking are denoted as **5x**, **10x** and **1x**, respectively. Figure 3 demonstrates the improvement per unit cost of each LLMs. It can be found that almost all three LLMs has the best improvement per unit cost in list-wise ranking, except text-davinci-002 on the Book dataset. Moreover, point-wise ranking and pair-wise ranking have similar improvement per unit cost. Although pair-wise ranking may achieve better performance than point-wise ranking in absolute metrics, the requirement to run multiple prompts for pair-wise ranking results in additional cost. Overall, we recommend to conduct list-wise ranking for recommendation tasks in practice, due to its decent performance and lower cost.

4.5 RQ4: Performance Under Different Shots Examples

Previous studies in NLP have emphasized that the number of examples M is important for in-context learning. To assess the impact of M in LLMs for recommendation, we conducted experiments on Movie dataset by varying M from 1 to 5. Figure 4 illustrates the performances of different M in terms of $NDCG@3$ of ChatGPT and GPT3.5s. Surprisingly, we observe that the best results did not always correspond to the maximum number of examples. One possible explanation is that while more example shots can provide more context and information for LLMs to understand the recommendation task, they may also introduce more noise, causing LLMs to learn unhelpful patterns. Therefore, the optimal number of prompt shots may depend on the specific LLM, task, and dataset.

4.6 Case Study of Exceptions

It is worth noting that the LLM may generate some invalid answers even under few-shot in-context learning, leading to a compliance rate may be less than 100%, as shown in Table 1. For instance, Table 4 highlights two exceptional cases of answers from ChatGPT with pair-wise ranking, where both cases lack a correct answer because